ZePRO
UG/PG Project Allocation System

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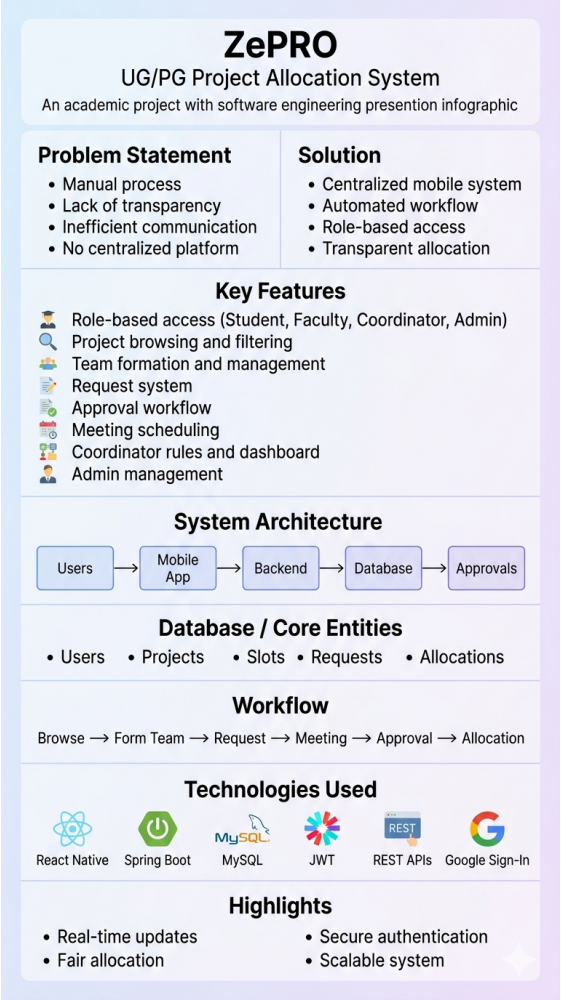
Introduction

- **ZePRO**: Centralized UG/PG project allocation system
- Students **explore and request** projects
- Faculty manage **slots and approvals**
- **Transparent** and efficient workflow



Problem Statement

The Current Scenario

Project allocation is currently handled through an informal and manual process, where students approach faculty individually without a centralized system.

Key Issues:

- **Time-Consuming:** Delays in identifying available faculty and getting approvals
- **Lack of Transparency:** No visibility of available project slots or faculty capacity
- **Inefficient Communication:** Repeated interactions between students and faculty
- **No Centralized Platform:** All activities are scattered and unorganized

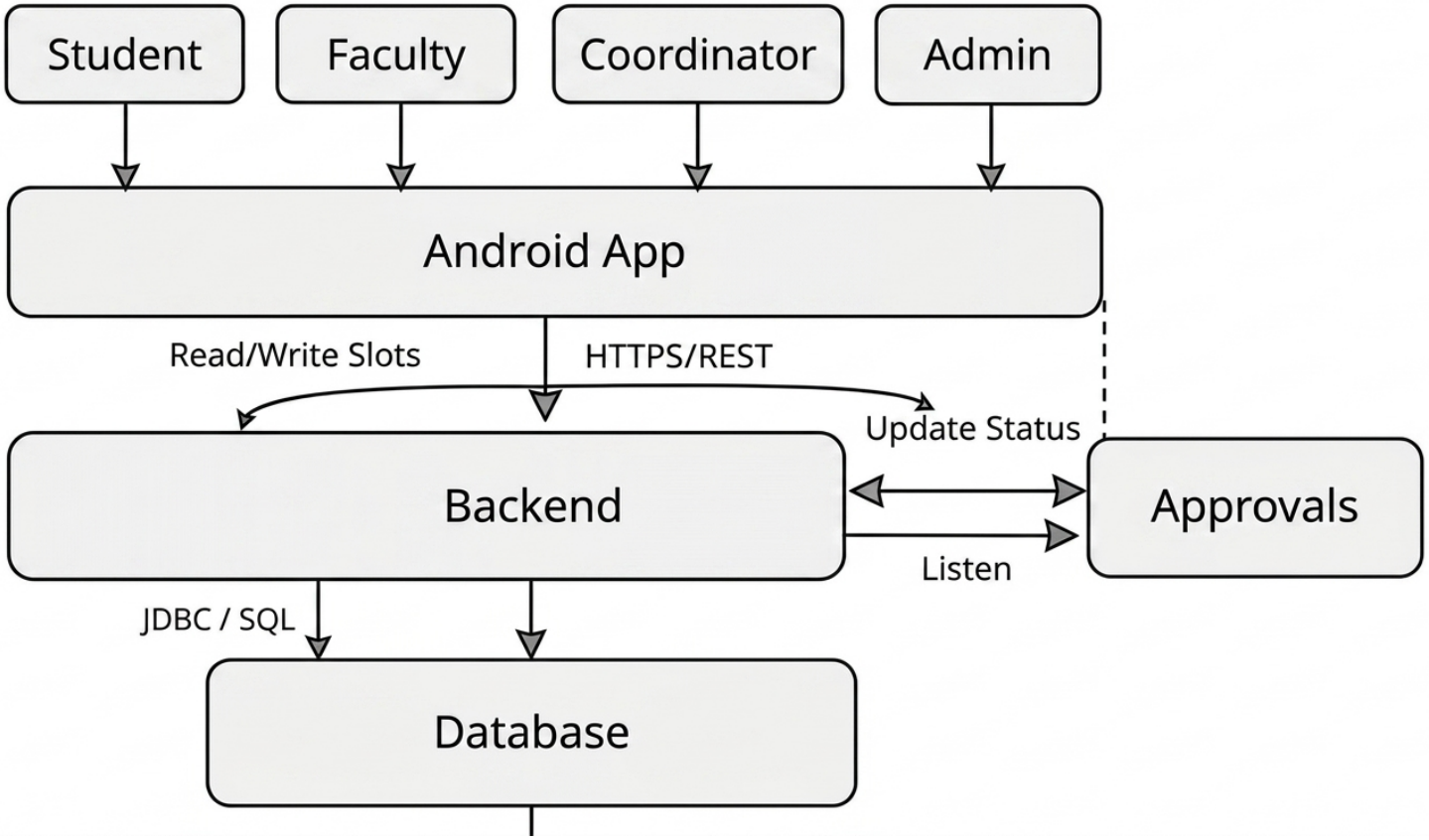
Objectives

- Automate the project allocation workflow
- Provide **centralized platform** for all users
- Ensure fair and transparent allocation
- Enable **role-based access** (Student, Faculty, Faculty Coordinator)
- Reduce **manual effort and delays**

Project Functionalities

- **User Authentication:** Secure role-based login (Student, Faculty, Coordinator)
- **Administration:** Manage institutes, departments, and coordinators
- **Project Management:** Faculty manage projects, domains, and slots
- **Project Selection:** Students browse and request projects
- **Team Formation:** Students form teams and submit requests
- **Approval Workflow:** Faculty accept, reject, or schedule meetings
- **Coordinator Control:** Manage allocation rules, slots, and deadlines

Overall System Design



- Mobile app handles **user interactions and requests**
- Backend ensures **secure processing and real-time updates**

Technology Stack & System Modeling

Tech Stack

- **Frontend:** React Native (Android)
- **Backend:** Spring Boot (REST APIs)
- **Database:** MySQL
- **Authentication:** JWT
- **Communication:** REST API (JSON)

System Modeling

- **Use Case:** Interaction between Student, Faculty, Coordinator, Admin
- **Workflow:** Request $\rightarrow$Meeting $\rightarrow$Approval $\rightarrow$ Allocation
- **Data Model:** Users, Projects, Slots, Requests, Allocations

Module-Wise Team Contributions

Team Member	Module & Responsibilities
Shaik Mehtab	**Student Module:** Create and manage teams, join teams, send project requests, browse and filter projects based on domain and faculty
Yangala Reddy Sekhar	**Faculty Module:** Create domains and sub-domains, add project topics, manage and update student slots, accept/reject teams, schedule and reschedule meetings
Konki Navadeep	**Faculty Coordinator Module:** Define allocation rules (team size, slots per faculty), manage deadlines, monitor dashboard, perform manual allocation, generate allocation reports (PDF)
Chilakala Nagendra Reddy	**Admin Module:** Create and manage institutions and departments, assign/remove faculty coordinators, maintain admin dashboard
PremChand	**Testing & Security:** Perform Appium testing, ensure secure authentication, validate workflows and system functionality

UI Snapshots

login_screen.png

faculty_dashboard.png

Code Snippet

```
if (facultySlots > 0) {
    approveRequest(teamId);
    facultySlots--;
    updateDatabase();
} else {
    rejectRequest(teamId);
}
```

Results and Conclusion

- Successfully digitized project allocation process
- Reduced manual workload
- Improved transparency and efficiency
- Real-time tracking of allocation status

Future Scope:

- Web version of application
- Advanced analytics dashboard
- Integration with colleg esystems